# Function
# Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Which of the following is true for the function $f(x) = 8x^3 - 7x$? 1

A. Odd function B. Even function

C. Neither odd nor even D. Zero function

$$f(x) = 8x^3 - 7x$$
$$f(-x) = 8 \times (-x)^3 - 7 \times (-x) = -(8x^3 - 7x) = -f(x)$$
$$\therefore \text{Odd function } f(-x) = -f(x)$$

2 Given $f(x) = x^2 - x$ and $h(x) = 2x + 1$, find:

a. $f(-2)$ 1

b. $f(h(x))$ 1

c. $x$ where $h(x) = 0$ 1

| | |
|---|---|
| (a) | $f(-2) = (-2)^2 - (-2) = 6$ |
| (b) | $f(h(x)) = f(2x+1)$ <br> $= (2x+1)^2 - (2x+1)$ <br> $= 4x^2 + 2x$ or $2x(2x+1)$ |
| (c) | $h(x) = 0$ <br> $2x + 1 = 0 \quad x = -\frac{1}{2}$ |

3 Find the equation of the line which is perpendicular to the line $3x - 5y - 7 = 0$ and which passes through the point (2, –6). 2

Find gradient of $3x - 5y - 7 = 0$
$$5y = 3x - 7$$
$$y = \frac{3}{5}x - \frac{7}{5}$$
Gradient $m_1 = \frac{3}{5}$

Gradient of perpendicular $m_2 = -\frac{5}{3}$

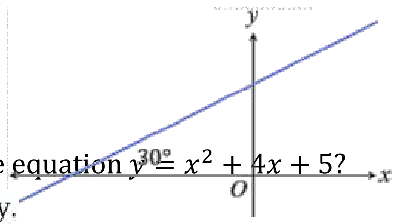
Equation $y - y_1 = m_2(x - x_1)$
$$y - -6 = -\frac{5}{3}(x - 2)$$
$$3y + 18 = -5x + 10$$
$$5x + 3y + 8 = 0 \quad OR \quad y = -\frac{5}{3}x - \frac{8}{3}$$

4 Find the radius and centre of the circle with the equation: 2

$$4x^2 - 4x + 4y^2 + 24y + 21 = 0$$

$$4x^2 - 4x + 4y^2 + 24y + 21 = 0$$
$$x^2 - x + y^2 + 6y = -\frac{21}{4}$$
$$\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + (y+3)^2 - 9 = -\frac{21}{4}$$
$$\left(x - \frac{1}{2}\right)^2 + (y+3)^2 = 4$$
$\therefore$ Radius is 2 and the centre $\left(\frac{1}{2}, -3\right)$

5 A line makes an angle of 30° with the positive direction of the $x$-axis as shown. What is the gradient of the line, in exact value. 1




$$m = \tan\theta$$
$$\therefore \tan 30°$$
$$1$$

6 What are the co... t of the graph with the equation $y = x^2 + 4x + 5$? 1

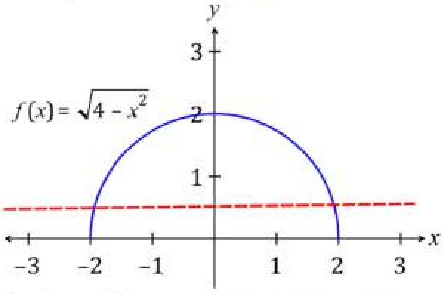
$\therefore$ Turning point occurs on the axis of symmetry.
$$x = -\frac{b}{2a} = -\frac{4}{2 \times 1} = -2$$
$$y = x^2 + 4x + 5$$

7 Which one of the following functions is not one-to-one? 1

A. $f(x) = 4 - x^2, x \geq 0$ B. $f(x) = \sqrt{4 - x^2}$

C. $f(x) = 1 - 4x$ D. $f(x) = \frac{4}{x}$

In a one-to-one function, every element in the range of a function corresponds to exactly one element of the domain.




Horizontal line test: intersects graph at two places.
$\therefore f(x) = \sqrt{4 - x^2}$ not a one-to-one function.

8 The equation $y = 2x^2 + 3x - 14$ represents a parabola on the number plane.

a. Find the intercepts of the parabola on the x-axis. 2

b. What is the equation of the axis of symmetry of the parabola? 1

a) $x$ intercept where $y = 0$

$$2x^2 + 3x - 14 = 0$$
$$2x^2 + 7x - 4x - 14 = 0$$
$$x(2x + 7) - 2(2x + 7) = 0$$
$$(2x + 7)(x - 2) = 0$$
$$2x = -7 \text{ or } x = 2$$
$$x = -3.5 \text{ or } x = 2$$

b) Axis is midway between the $x$ intercepts

$$x = \frac{-3.5 + 2}{2} = \frac{-1.5}{2} = -0.75$$

9 What is the solution to the equation $6x^2 = x + 2$? 1

$$6x^2 - x - 2 = 0$$
$$(3x - 2)(2x + 1) = 0$$
$$(3x - 2) = 0 \text{ or } (2x + 1) = 0$$
$$\therefore x = \frac{2}{3} \text{ or } x = -\frac{1}{2}$$

10 $A(3, -1)$ is a vertex of the rectangle $ABCD$ whose side $AD$ has equation $2x - y - 7 = 0$. Find in general form the equation of the side $AB$. 2

$AB \perp AD$ where $AD$ is the line $y = 2x - 7$ with gradient 2. Hence $AB$ has gradient $-\frac{1}{2}$.

Equation of $AB$ is
$$y + 1 = -\tfrac{1}{2}(x - 3)$$
$$2y + 2 = -x + 3$$
$$x + 2y - 1 = 0$$

11 Given that $f(x) = x^2 - x$ and $h(x) = 2x + 1$, find:

(a) $f(-2) = (-2)^2 - (-2) = 6$

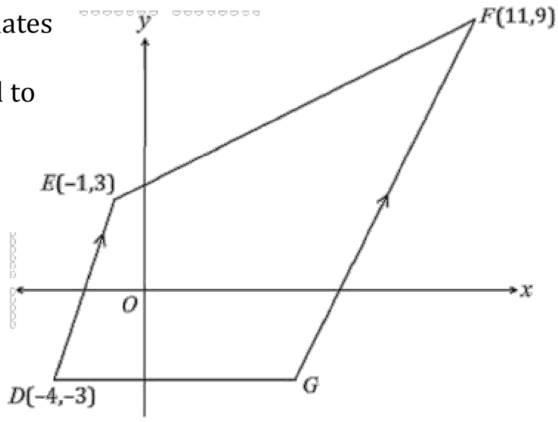
b) $f(h(x)) = f(2x + 1)$
$$= (2x + 1)^2 - (2x + 1)$$
$$= 4x^2 + 2x \text{ or } 2x(2x + 1)$$

a. $f(-2)$ 1

b. $f(h(x))$ 1

c. $x$ when $h(x) = 0$. 1

12 On a number plane the points $D$, $E$ and $F$ have coordinates $(-4, -3)$, $(-1, 3)$ and $(11, 9)$ respectively. $DE$ is parallel to $GF$ and $DG$ is parallel to the $x$-axis.




a. What is the gradient of DE? 1

b. What is the midpoint of DE? 1

c. Find the equation of the line FG. 2

d. Show that the coordinates of the point G are $(5, -3)$. 2

e. Find the distance EF. 1

f. Find the equation of the circle centred at F with radius EF. 1

| | |
|---|---|
| (a) | $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-(-3)}{-1-(-4)} = \frac{6}{3} = 2$ |
| (b) | $x = \frac{x_1 + x_2}{2} = \frac{-4+(-1)}{2} = -\frac{5}{2}$<br>$y = \frac{y_1 + y_2}{2} = \frac{-3+3}{2} = 0$<br>$\therefore$ Midpoint is $(-2.5, 0)$ |
| (c) | Line $FG$ has the same gradient as $DE$ (parallel lines)<br>$y - y_1 = m(x - x_1)$<br>$y - 9 = 2(x - 11)$<br>$y - 9 = 2x - 22$<br>$2x - y - 13 = 0$ |
| (d) | Point $G$ is on the line $DG$ or $y = -3$<br>To find $x$ when $y = -3$ substitute into the equation of $FG$<br>$2x - (-3) - 13 = 0$<br>$2x = 10 \quad x = 5$<br>$\therefore$ Coordinates of $G$ are $(5, -3)$ |
| (e) | Use Pythagoras theorem<br>$EF^2 = 12^2 + 6^2$<br>$EF = \sqrt{12^2 + 6^2} = \sqrt{180} = 3\sqrt{20}$ |
| (f) | $(x-a)^2 + (y-b)^2 = r^2$<br>$(x-11)^2 + (y-9)^2 = (\sqrt{180})^2$<br>$(x-11)^2 + (y-9)^2 = 180$ |

13 The function $f(x) = 4 - 3x$ has the given domain $-2 < x \leq 3$. What is its range? 1

Function is continuously decreasing within the domain, range extends between the end values.

$f(-2) = 10$, excluded $\quad f(3) = -5$, included

14 What is the domain and range of the function $f(x) = \sqrt{2-x}$? 1

D: $2 - x \geq 0$
$x \leq 2$
$(-\infty, 2]$

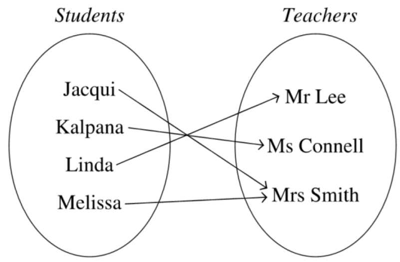
R: $y \geq 0$
$[0, \infty)$

# Function
# Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 The diagram shows the relation of Year 11 students and their Mathematics teachers. Which of the following most accurately describes the relation shown? 1




One - to - many

2 Consider the function $f(x) = x^{\frac{2}{3}}$.

a. Evaluate $f(8)$.

b. Simplify $f(a^2) \times f(a^{-1})$.

c. State if $f(x)$ is even, odd or neither. Justify your answer.

d. Let $g(x) = (3x-8)^{\frac{3}{4}}$. Find $f\big(g(x)\big)$ and state its domain.

**a** $8^{\frac{2}{3}} = \left(8^{\frac{1}{3}}\right)^2 = 2^2 = 4$ 1

**b** $(a^2)^{\frac{2}{3}} \times (a^{-1})^{\frac{2}{3}}$ 1

$= a^{\frac{4}{3}} \times a^{-\frac{2}{3}} = a^{\frac{2}{3}}$ or $\sqrt[3]{a^2}$ 1

**c** $f(-x) = (-x)^{\frac{2}{3}} = \sqrt[3]{(-x)^2}$ 2

$= \sqrt[3]{x^2} = x^{\frac{2}{3}} = f(x)$

Even

**d** $f(g(x)) = [g(x)]^{\frac{2}{3}}$

$= \left[(3x-8)^{\frac{3}{4}}\right]^{\frac{2}{3}}$

$= (3x-8)^{\frac{1}{2}}$ or $\sqrt{3x-8}$

Domain: $x \geq \dfrac{8}{3}$

3 The function $h$ is given that $h(x) = -\sqrt{4-x^2}$.

a. What is the domain and range of $h$? 1

Domain : $-2 \leq x \leq 2$ or $[-2, 2]$

Range : $-2 \leq y \leq 0$ or $[-2, 0]$

b. Show algebraically if $h$ is an even function, an odd function, or neither even nor odd. 2

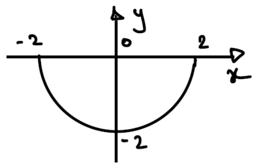
$h(x) = -\sqrt{4-x^2}$

$h(-x) = -\sqrt{4-(-x)^2} = -\sqrt{4-x^2} = h(x)$

Since $h(x) = h(-x)$

$\therefore$ $h(x)$ is an even function.

c. Sketch a graph of the function, showing important features and intercepts. 1




d. Is $h$ one-to-one or many-to-one? Explain your reasoning. 2

$h(x)$ is many-to-one as when we draw a horizontal line, it cuts the graph twice meaning two $x$ values give the same $y$ value.

4 Points $P(3,4)$, $Q(1,2)$ and $R(a,b)$ are collinear. Which of the following statements is true? 1

A. $a-b=1$ B. $a-b=-1$

C. $a-b=3$ D. $a-b=-3$

$m_{PQ} = \dfrac{2-4}{1-3} = \dfrac{-2}{-2} = 1$

$m_{QR} = \dfrac{b-2}{a-1} = 1$

$b-2 = a-1$

$a-b = -1$

5 A line $l$ passes through the points $(2, 0)$ and $(8, 3)$. What is the gradient of $l$? 1

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{8-2} = \frac{3}{6}$$

6 Which of the following is an even function? 1

A. $sinx$ B. $x^3 + x$

C. $x^6 + x^4 + x$ D. $x^6 + x^4 + x^2$

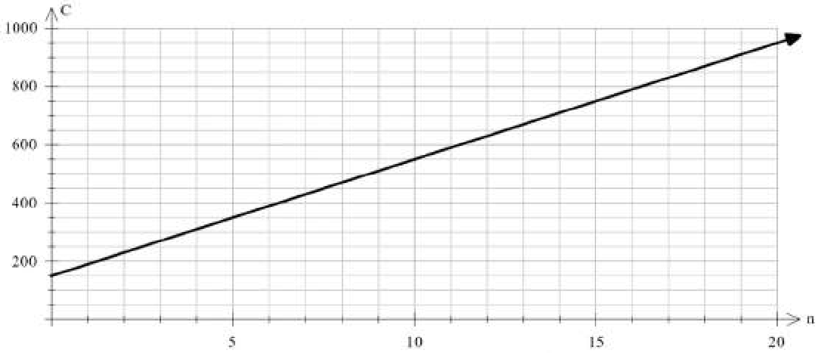
$f(x) = x^6 + x^4 + x^2$

$f(-x) = (-x)^6 + (-x)^4 + (-x)^2$

$= x^6 + x^4 + x^2$

$f(-x) = f(x)$ $\therefore$ (D)

7 The graph below shows the cost $C$, of producing n widgets for a company manufacturing widgets in Western Sydney.




a. What is the cost to the company of manufacturing widgets if none are produced this week due to a Covid-19 outbreak in Sydney? 1

b. Write an equation for the cost of manufacturing widgets in terms of $C$ and $n$. 1

c. The company sells each widget for \$50. Draw a revenue graph to represent sales of widgets on the grid above. 1

d. What is the break-even point for sales of widgets? Explain what this means? 2

a). \$150

b). $C = mn + b$

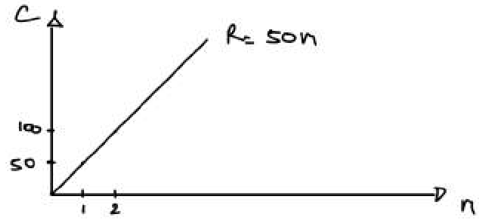
$b$: 150

$m = \frac{rise}{run}$

$= \frac{350 - 150}{5} = 40$

$\therefore \boxed{C = 40n + 150}$

c).




d). Breakeven : $C = R$

$50n = 40n + 150$

$10n = 150$

$n = 15$ widgets

When the company sells 15 widgets, it will make no profit or loss.

8 Which of the following functions has a discontinuity? 1

A. $f(x) = |x|$ B. $f(x) = x^2$

C. $f(x) = \frac{x}{x^2+1}$ D. $f(x) = \frac{|x|}{x}$

Only the function $f(x) = \frac{|x|}{x}$ has a domain which does not include all real numbers. This function is not defined at $x = 0$, and as such has a discontinuity here.

9 a. Factorise $9x^3 + 3x^2 - 12x - 4$. 1

$9x^3 + 3x^2 - 12x - 4 = 3x^2(3x+1) - 4(3x+1) = \boxed{(3x^2-4)(3x+1)}$

b. Hence, solve $18x^4 + 6x^3 - 24x^2 - 8x = 0$. 2

$18x^4 + 6x^3 - 24x^2 - 8x = 0$

$2x(9x^3 + 3x^2 - 12x - 4) = 0$

$2x(3x^2 - 4)(3x+1) = 0$

$2x = 0, \quad 3x^2 - 4 = 0, \quad 3x + 1 = 0$

$x = 0, \quad x^2 = \frac{4}{3}, \quad x = -\frac{1}{3}$

$x = \pm\frac{2}{\sqrt{3}}$

$\therefore \boxed{x = 0, \pm\frac{2}{\sqrt{3}}, -\frac{1}{3}}$

10 What is the domain of the function $f(x) = \frac{x}{\sqrt{x+5}}$? 1

A. $(-5, \infty)$ B. $[-5, \infty)$

C. $(-\infty, -5)$ D. $(-\infty, -5]$

$x + 5 > 0$ (square root can't be negative, and denominator can't equal 0.) $x > -5$

In interval notation: $(-5, \infty)$

11 Find the points of intersection of $y = 3x - 6$ and $y = x^2 - 2x - 2$. 2

We can use the substitution method and set the equations equal to each other:

$3x - 6 = x^2 - 2x - 2$ (move $3x$ and $-6$ terms to right-hand side)

$x^2 - 5x + 4 = 0$ (factorise)

$(x-1)(x-4) = 0$

$x = 1$ or $x = 4$

Substitute $x$-values into one of the given equations:

When $x = 1$, $y = 3(1) - 6 = -3$

When $x = 4$, $y = 3(4) - 6 = 6$

Intersections: $(1, -3)$, $(4, 6)$

12 Complete the square to solve (give the answer in surd form): $z^2 - z = 1$ 1

$z^2 - z = 1$

$z^2 - z + \frac{1}{4} = 1 + \frac{1}{4}$

$\left(z - \frac{1}{2}\right)^2 = \frac{5}{4}$

$z - \frac{1}{2} = \pm\sqrt{5/4}$

$z = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$

13 Use the quadratic formula to solve the equation: $5x^2 + 3x - 1 = 0$ 1

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-3 \pm \sqrt{9 - 4(5)(-1)}}{2 \cdot 5} = \frac{-3 \pm \sqrt{29}}{10}$

14 Sketch the function $y = x^2 + 4x - 5$, including the vertex and intercepts. 1

$y = x^2 + 4x - 5 = (x+5)(x-1)$

X-INTS $x = -5 \quad x = 1$

Y-INT. $y = -5$

$V_x = \frac{-b}{2a} = \frac{-4}{2} = -2$

$V_y = (-2+5)(-2-1) = -9$

$\therefore V(-2, -9)$